Hence acceleration is diagonally down and to the left.

5 **C** The platform-man system is in equilibrium, i.e. resultant force = zero.

Weight of platform-man = Force by water on platform-man

By Newton's 3rd Law, |Force by water on platform-man| = |Force by platform-man on water|

By Newton's 2nd Law, Force by platform-man on water = $\frac{\Delta p}{\Delta t} = \frac{\Delta (mv)}{\Delta t} = \frac{\Delta m}{\Delta t} \times v = 40v$ (where v is the velocity of water leaving the platform).

$mg = 40v$

$v = \frac{(96)(9.81)}{40} = 23.54 = 24 \text{ m s}^{-1}$ (2 s.f)

6 **C** Impulse = change in momentum = final momentum – initial momentum

Taking the direction away from the player as the positive direction